

CS5275 Problem Set 1

Li Jiaru (Student ID: A0332008U)

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Problem 1

- a. Pick a non-leaf vertex $v \in V$. Removing v from T splits it into a set of (at least two) connected components $\{T_1, T_2, \dots\}$. If $|T_i| > n/2$ for some T_i , we move v to its neighbour in T_i and repeat until every T_i satisfies $|T_i| \leq n/2$.

Now we build a set X with $n/4 \leq |X| \leq n/2$ via the following procedure:

- If $n/4 \leq |T_i| \leq n/2$ for some T_i , take $X := T_i$.
- Otherwise, every T_i satisfies $|T_i| < n/4$. Initialise $X \leftarrow \emptyset$. While $|X| < n/4$, we union X with some T_i , i.e., $X \leftarrow X \cup T_i$. Then we are guaranteed to get some X that satisfies the requirement (similar to the argument shown in Lecture 1, Page 18).

Note that $\partial_{\text{out}}(X)$ consists of only v itself as it splits T into X and $T \setminus X$. Thus we have $|\partial_{\text{out}}(X)| = 1$, so by definition

$$h_{\text{out}}(T) = \min_{0 < |S| \leq \frac{n}{2}} \frac{|\partial_{\text{out}}(S)|}{|S|} \leq \frac{|\partial_{\text{out}}(X)|}{|X|} = \frac{1}{|X|} \leq \frac{4}{n},$$

and so $h_{\text{out}}(T) \in O(1/n)$ as required.

- b. For any set $S \subset T$ with $0 < |S| \leq n/2$, there must be at least one edge between S and $T \setminus S$, as T is connected. Thus $|\partial_{\text{out}}(S)| \geq 1$ and

$$h_{\text{out}}(T) = \min_{0 < |S| \leq \frac{n}{2}} \frac{|\partial_{\text{out}}(S)|}{|S|} \geq \frac{1}{|S|} \geq \frac{2}{n},$$

and so $h_{\text{out}}(T) \in \Omega(1/n)$ as required.

Problem 2

- a. In the given procedure, at step i we denote the current set as W_i and subset trimmed as S_i . We also denote the union of all the subsets trimmed as $S := V \setminus W_{\text{final}}$.

Using the hint, to prove that S yields a low-conductance cut in G , we first try to bound $|E_G(S, W_{\text{final}})|$, the number of edges between S and W_{final} .

At step i , we have $\text{vol}_{G[W_i]}(S_i) \leq \text{vol}_{G[W_i]}(W_i)/2$ and $\Phi_{G[W_i]}(S_i) \leq \phi/2$, so by definition

$$|E_{G[W_i]}(S_i, W_i \setminus S_i)| \leq \text{vol}_{G[W_i]}(S_i) \frac{\phi}{2}.$$

Note that by definition

$$E_G(S, W_{\text{final}}) = \bigcup_i E_{G[W_i]}(S_i, W_i \setminus S_i) \cap E_G(S_i, W_{\text{final}}),$$

so that

$$|E_G(S, W_{\text{final}})| \leq \sum_i |E_{G[W_i]}(S_i, W_i \setminus S_i)| \leq \sum_i \frac{\phi}{2} \text{vol}_{G[W_i]}(S_i).$$

As $G[W_i]$ is a subgraph of G and $S = \bigcup_i S_i$,

$$\text{vol}_{G[W_i]}(S_i) \leq \text{vol}_G(S_i), \quad \sum_i \text{vol}_G(S_i) = \text{vol}_G(S),$$

so we get

$$|E_G(S, W_{\text{final}})| \leq \frac{\phi}{2} \text{vol}_G(S).$$

Now we follow the hint to restrict this cut to G' . Denote $S' := S \cup V'$ and $W' = W_{\text{final}} \cup V'$. Then we have

$$|E_{G'}(S', W')| \leq |E_G(S, W_{\text{final}})| \leq \frac{\phi}{2} \text{vol}_G(S).$$

As $\Phi(G') \geq \phi$, we have

$$|E_{G'}(S', W')| \geq \phi \min(\text{vol}_{G'}(S'), \text{vol}_{G'}(W')).$$

Combining these two inequalities gives

$$\min(\text{vol}_{G'}(S'), \text{vol}_{G'}(W')) \leq \frac{1}{2} \text{vol}_G(S).$$

Since $\text{vol}_{G[W_i]}(S_i) \leq \text{vol}_{G[W_i]}(W_i)/2$ at every step i , we must have $\text{vol}_{G'}(S') \leq \text{vol}_{G'}(W')$, so that $\min(\text{vol}_{G'}(S'), \text{vol}_{G'}(W')) = \text{vol}_{G'}(S')$ and

$$\text{vol}_{G'}(S') \leq \frac{1}{2} \text{vol}_G(S).$$

As $|E'| \geq (1 - \varepsilon)|E|$ and $2|E| = \text{vol}_G(V)$, we have

$$\text{vol}_G(S) - \text{vol}_{G'}(S') \leq 2|E| - 2|E'| \leq 2|E| - 2(1 - \varepsilon)|E| = 2\varepsilon|E| = \varepsilon \text{vol}_G(V).$$

Combining these two inequalities gives

$$\text{vol}_G(S) \leq 2\varepsilon \text{vol}_G(V).$$

Finally, as $S = V \setminus W_{\text{final}}$, this means

$$\text{vol}_G(W_{\text{final}}) \geq (1 - 2\varepsilon) \text{vol}_G(V) \in O(1 - \varepsilon) \text{vol}_G(V),$$

as required.

Problem 3

- a. Consider any vertex v . By definition it must have r neighbours, each of which has $r - 1$ new neighbours in turn. Thus the number of vertices at distance i from v is $r(r - 1)^{i-1}$ for $i \geq 1$ (via BFS). Let $B(v, t)$ be the ball of radius t around v . Then for some sufficiently large C , the total size of the ball is

$$\begin{aligned} |B(v, t)| &= 1 + \sum_{i=1}^t r(r-1)^{i-1} = 1 + r \sum_{i=0}^{t-1} (r-1)^i \\ &= \begin{cases} \frac{r(r-1)^t - 2}{r-2}, & r \neq 2, \\ 1 + 2t, & r = 2 \end{cases} \\ &\leq C(r-1)^t. \end{aligned}$$

As $r \in O(1)$, take $k := \left\lfloor \frac{\log_{r-1} n}{2} \right\rfloor \in \Theta(\log n)$, so that

$$|B(v, k)| \leq C(r-1)^k \leq C(r-1)^{\frac{\log_{r-1} n}{2}} = C\sqrt{n}.$$

Hence the number of pairs $(u, v) \in V \times V$ with $\text{dist}_G(u, v) \leq k$ is at most

$$\sum_{v \in V} |B(v, k)| = n|B(v, k)| \leq Cn^{3/2},$$

and so there exist $n^2 - Cn^{3/2} \in \Omega(n^2)$ pairs satisfying $\text{dist}_G(u, v) \in \Omega(\log n)(> k)$, as required.

- b. By definition, we have

$$\text{LR}(G, H) = \frac{\sum_{u,v} w_G(u, v) d_{\text{OPT}}(u, v)}{\sum_{u,v} w_H(u, v) d_{\text{OPT}}(u, v)} \leq \frac{\sum_{u,v} w_G(u, v) d(u, v)}{\sum_{u,v} w_H(u, v) d(u, v)},$$

where d_{OPT} is an optimal metric of the corresponding LP and $d(u, v) := \text{dist}_G(u, v)$ is the shortest-path metric on G .

For the numerator, notice that $w_G(u, v) = 1$ when $(u, v) \in E_G$ and 0 otherwise. But when $(u, v) \in E_G$, by definition $d(u, v) = 1$. Thus it equals to the number of edges, which is $rn/2 \in \Theta(n)$ as G is r -regular.

For the denominator, notice that $w_H(u, v) = 1$ for all $(u, v) \in E_H$ as H is the clique K_n . By (3.a), there exist $\Omega(n^2)$ pairs satisfying $d(u, v) \in \Omega(\log n)$, so it is $\sum_{u,v} d(u, v) \in \Omega(n^2 \log n)$.

Hence we have

$$\text{LR}(G, H) \in \frac{\Theta(n)}{\Omega(n^2 \log n)} = O\left(\frac{1}{n \log n}\right). \quad (1)$$

Consider the generalised conductance $\Phi(G, H)$ defined by

$$\Phi(G, H) = \min_{S: E_H(S, V \setminus S) \neq \emptyset} \frac{|E_G(S, V \setminus S)|}{|E_H(S, V \setminus S)|}.$$

Take the set S that minimises $\Phi(G, H)$. WLOG we assume that $|S| \leq n/2$, so that $|V \setminus S| \geq n/2$.

As G is r -regular, we have $\min(\text{vol}_G(S), \text{vol}_G(V \setminus S)) = \text{vol}_G(S) = r|S|$. Therefore $|E_G(S, V \setminus S)| = \frac{r|S|}{\Phi(G)}$.

As H is a clique, we have $|E_H(S, V \setminus S)| = |S| \cdot |V \setminus S|$.

Putting them together, we have

$$\Phi(G, H) = \frac{|E_G(S, V \setminus S)|}{|E_H(S, V \setminus S)|} = \frac{r|S|}{|S| \cdot |V \setminus S| \Phi(G)} \leq \frac{2r}{n\Phi(G)} \in O\left(\frac{1}{n}\right), \quad (2)$$

as $r \in O(1)$ and $\Phi(G) \in \Omega(1)$ as given.

Combining (1) and (2) gives

$$\text{LR}(G, H) \in O\left(\frac{\Phi(G, H)}{\log n}\right),$$

as required.

Problem 4

- a. We show that the path graph $G := P_n$ has the desired property. Let x and y be the two endpoints. Thus $\text{dist}_G(x, y) = n - 1 \in O(n)$.

If $A = \emptyset$, then $d_A(x, y) = 0$, so it does not affect the expectation. Thus for any nonempty A , by the triangle inequality and linearity of expectation, we have

$$\mathbb{E}[d_A(x, y)] = \mathbb{E}[|f_A(x) - f_A(y)|] \leq \mathbb{E}[f_A(x)] + \mathbb{E}[f_A(y)].$$

Consider $f_A(x) = \text{dist}_G(x, A)$. As G is a path graph, the event $f_A(x) > k$ is equivalent to $v_i \notin A$ for all $i = 1, 2, \dots, k$. Since each vertex is independently sampled with probability $p = 1/2$, we have $\Pr[v_i \in A] = 1/2$ and so

$$\Pr[f_A(x) > k] = \prod_{i=1}^k \Pr[v_i \notin A] = \left(1 - \frac{1}{2}\right)^k = \frac{1}{2^k}.$$

By the tail-sum formula,

$$\mathbb{E}[f_A(x)] = \sum_{k=0}^n \Pr[f_A(x) > k] = \sum_{k=0}^n \frac{1}{2^k} < \sum_{k=0}^{\infty} \frac{1}{2^k} = 2,$$

and by symmetry we have $\mathbb{E}[f_A(y)] < 2$ as well. Thus $\mathbb{E}[d_A(x, y)] < 4$.

Therefore

$$\mathbb{E}[d_A(x, y)] \in O(1) = O(n^{-1})O(n) \subset O(n^{-0.01}) \text{dist}_G(x, y)$$

as required.

- b. We show that the star graph $G := S_n$ has the desired property. Let x and y be two distinct leaves.

For any nonempty A , if $x, y \notin A$, then $f_A(x) = f_A(y)$ and so $d_A(x, y) = |f_A(x) - f_A(y)| = 0$. Thus $d_A(x, y) \neq 0$ only when $x \in A$ or $y \in A$. Note that $\Pr[x \in A] = \Pr[y \in A] = p = 1/n$. By the inclusion-exclusion principle, we have

$$\begin{aligned} \Pr[x \in A \cup y \in A] &= \Pr[x \in A] + \Pr[y \in A] - \Pr[x \in A \cap y \in A] \\ &\leq \Pr[x \in A] + \Pr[y \in A] \\ &= \frac{1}{n} + \frac{1}{n} = \frac{2}{n}. \end{aligned}$$

Note that $d_A(x, y) \leq \text{dist}_G(x, y)$ for all A . Hence the expectation

$$\begin{aligned} \mathbb{E}[d_A(x, y)] &\leq \Pr[d_A(x, y) \neq 0] \max(d_A(x, y)) \\ &\leq \frac{2}{n} \text{dist}_G(x, y) \\ &\in O(n^{-0.01}) \text{dist}_G(x, y), \end{aligned}$$

as required.